Two matrices with identical central zonotopal algebras but different projective configurations

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Abstract

The central matrix-algebra conjecture in MathDB 338791 is false. Two explicit bridge-free 2×4 integer matrices have identical central zonotopal ideals, but their column configurations have disjoint cross-ratio orbits and hence are not z-equivalent.

Task 1. The original conjecture

For a full-row-rank real matrix A with columns $v_1, \dots, v_m$, put $Z_A = \sum_i [0, v_i]$. For each facet H of Z_A, choose a nonzero normal η_H, and let $m(H)$ count the nonzero entries of $\eta_H A$. The central zonotopal algebra is

$$C_A^{\mathcal{C}} = \mathbb{R}[x_1, \dots, x_n]/I_A^{\mathcal{C}}, \qquad I_A^{\mathcal{C}} = \langle (\eta_H \cdot x)^{m(H)} : H \text{ a facet} \rangle.$$

Two matrices are z-equivalent if their row spaces differ by an invertible diagonal coordinate change and a coordinate permutation. A bridge-column is a column whose deletion lowers the rank. The conjecture asserts that isomorphism of central algebras, graded or ungraded, is equivalent to z-equivalence after removing bridge-columns and corresponding dependent rows. This is Conjecture 2 in the preprint and Conjecture 16 in the final article [1]. The following examples have no bridge-columns, so no reduction is needed.

Proposition 1. *For $t \in \mathbb{R} \setminus \{0, 1\}$ let*

$$A_t = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & t \end{pmatrix}.$$

Then $I_{A_t}^{\mathcal{C}} = (x, y)^3$. Nevertheless, A_2 and A_3 are not z-equivalent. Thus they disprove the conjecture.

Proof. Every two columns of A_t are independent. Hence its rank is two and deleting any single column leaves rank two. Its zonotope has four pairs of parallel edges, parallel to its four distinct column directions. Indeed, for any normal η, the face maximizing $\eta \cdot z$ fixes an endpoint of each segment with $\eta \cdot v_i \neq 0$, and leaves free the segments with $\eta \cdot v_i = 0$. Such a face is an edge exactly when one column direction is perpendicular to η.

We may therefore use the four normal directions $(1, 0), (0, 1), (1, -1), (t, -1)$. Each normal annihilates exactly one column, so all four exponents are three. Consequently

$$I_{A_t}^{\mathcal{C}} = \langle x^3, y^3, (x - y)^3, (tx - y)^3 \rangle. \tag{1}$$

Modulo $\langle x^3, y^3\rangle$, the last two generators have coefficient matrix, in the basis x^2y, xy^2,

$$\begin{pmatrix} -3 & 3 \\ -3t^2 & 3t \end{pmatrix}, \qquad \det = 9t(t-1) \neq 0.$$

They therefore generate both mixed cubic monomials as well. All four generators in (1) are homogeneous cubics, proving equality with $(x, y)^3$. In particular both algebras are literally

$$\mathbb{R}[x, y]/(x, y)^3,$$

with graded basis $1, x, y, x^2, xy, y^2$ and Hilbert series $1 + 2q + 3q^2$.

It remains to rule out z-equivalence. For four nonproportional vectors $w_1, \ldots, w_4 \in \mathbb{R}^2$, define

$$r(w_1, w_2, w_3, w_4) = \frac{\det(w_1, w_3)\det(w_2, w_4)}{\det(w_1, w_4)\det(w_2, w_3)}.$$

This number is unchanged by a common invertible linear transformation or by independent nonzero rescalings of the vectors: the determinant factors cancel. Permuting the vectors gives the set

$$\left\{ r, \frac{1}{r}, 1-r, \frac{1}{1-r}, \frac{r}{r-1}, \frac{r-1}{r} \right\}. \tag{2}$$

For completeness, adjacent transpositions change the ratio by $r \mapsto 1/r$ or $r \mapsto 1-r$; these generate precisely the six displayed expressions. Coincidences can reduce the number of distinct values. For the columns of A_t in their displayed order, the ratio is $1/t$. Thus the two permutation orbits are

$$\mathcal{R}(A_2) = \{-1, \tfrac{1}{2}, 2\}, \qquad \mathcal{R}(A_3) = \{-2, -\tfrac{1}{2}, \tfrac{1}{3}, \tfrac{2}{3}, \tfrac{3}{2}, 3\}.$$

They are disjoint. If the row spaces of A_2 and a column-permuted and column-rescaled A_3 were equal, their two row bases would differ by an invertible 2×2 matrix. This would give projectively equivalent column configurations, contradicting the disjoint orbits. Both algebra isomorphism conditions hold, while the third conjectured condition fails. □

The same computation gives infinitely many inequivalent realizations with this one central algebra. In fact, for $s, t > 2$, (2) contains just one value greater than two, namely the parameter itself; hence $A_s \sim_z A_t$ forces $s = t$. All these matrices represent the same uniform matroid $U_{2,4}$. The result concerns arbitrary real matrices and does not resolve the separate conjecture for graphical or totally unimodular configurations.

Verification and scope

The accompanying exact verifier checks every column deletion, enumerates the zonotope vertices and facet directions, computes the full cubic coefficient matrices, and evaluates all 24 column orderings. As a separate check, it rules out all 24 projective matchings by rational linear algebra, with self-equivalences as positive controls. Computation supports the explicit proof; no numerical approximation is needed. The source's displayed square dimension for the reduced matrix is inconsistent with the stated deletion operation. Here that operation deletes nothing, so the counterexample addresses its intended meaning without using that typo. The proof was checked by its authoring agent, without independent peer review or a proof assistant.

Four follow-up tasks

These are further questions, not additional campaign credits or claims of new open problems.

Task 2. Classify the central algebras of rank-two configurations with repeated column directions, in terms of their multiplicities and projective parameters.

Task 3. Determine for each uniform matroid $U_{n,m}$ whether all real realizations have the same central algebra, and describe that algebra.

Task 4. Resolve the separate graphical conjecture: does the central algebra determine the graphical matroid after bridge deletion?

Task 5. Find a natural additional structure on the central algebra that recovers the projective column configuration, including its cross ratios.

References

[1] G. Nenashev, *Classification of external Zonotopal algebras*, Electronic Journal of Combinatorics 26(1) (2019), P1.32, doi:10.37236/8299; preprint arXiv:1803.09966v1.